

The conjugacy of fractional stochastic differential equations and random differential equations

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Fractional Brownian motions

Fractional Brownian motion (fBm) is a family of centered Gaussian processes $B^H = \{B^H(t)\}$, $t \in \mathbb{R}$ or $\mathbb{R}_+$, indexed by the Hurst parameter $H \in (0, 1)$ with continuous sample paths and the covariance function

$$R_H(s, t) = \frac{1}{2}(t^{2H} + s^{2H} - |t - s|^{2H}).$$

Mandelbrot and van Ness (1968): B^H is

- ▶ self-similar: $B^H(at) \sim |a|^H B^H(t)$,
- ▶ stationary increments: $B^H(t) - B^H(s) \sim B^H(t - s)$,
- ▶ long memory when $H > \frac{1}{2}$: $\sum_{n=1}^{\infty} E[B^H(1)(B^H(n+1) - B^H(n))] = \infty$,
- ▶ non-Markovian,
- ▶ not semi-martingale if $H \neq \frac{1}{2}$: can not construct stochastic integral w.r.t. fBm in the sense of probability convergence of sequence of Darboux sum,
- ▶ almost no where differentiable, but almost sure Hoelder continuous: for every $\epsilon > 0$,

Rough path theory for fractional stochastic integrals

$$C^\nu([0, T], \mathbb{R}^d) = \left\{ f : [0, T] \rightarrow \mathbb{R}^d, \|f\|_{\infty, \nu, 0, T} < \infty \right\}$$

$$\|f\|_{\infty, \nu, 0, T} := \|f\|_{\infty, 0, T} + \|f\|_{\nu, 0, T} = \|f\|_{\infty, 0, T} + \sup_{0 \leq s < t \leq T} \frac{\|f(t) - f(s)\|}{|t - s|^\nu}$$

$$\begin{aligned} W_0^{\alpha, 1}([0, T], \mathbb{R}^d) &= \left\{ f : [0, T] \rightarrow \mathbb{R}^d, \|f\|_{\alpha, 1} \right. \\ &:= \left. \int_0^T \left(\frac{\|f(s)\|}{s^\alpha} + \int_0^s \frac{\|f(s) - f(u)\|}{|s - u|^{1+\alpha}} du \right) ds < \infty \right\} \end{aligned}$$

$$W_T^{1-\alpha, \infty}([0, T], \mathbb{R}^d) = \left\{ g : [0, T] \rightarrow \mathbb{R}^d, \|g\|_{1-\alpha, \infty, T} < \infty \right\}$$

$$\|g\|_{1-\alpha, \infty, T} := \sup_{0 \leq s < t \leq T} \left(\frac{\|g(t) - g(s)\|}{|t - s|^{1-\alpha}} + \int_s^t \frac{\|g(u) - g(s)\|}{|u - s|^{2-\alpha}} du \right).$$

For each fixed $\frac{1}{2} < \nu < H$, choose $\alpha \in (1 - \nu, \frac{1}{2})$. It can be proved (see Nualart and Răşcanu (2002)) that

$$C^\nu([0, T], \mathbb{R}^d) \subset W_T^{1-\alpha, \infty}([0, T], \mathbb{R}^d) \subset C^{1-\alpha}([0, T], \mathbb{R}^d) \subset W_0^{\alpha, 1}([0, T], \mathbb{R}^d).$$

Since B^H is an fBm, each trajectory $B^H(\cdot, \omega) = \omega(\cdot)$ belongs to $C^\nu([0, T], \mathbb{R})$. Thus for each $f \in W_0^{\alpha,1}([0, T], \mathbb{R}^d)$ and each trajectory $\omega \in W_T^{1-\alpha,\infty}([0, T], \mathbb{R})$ we can define fractional derivatives

$$D_{a+}^\alpha f(s) := \frac{1}{\Gamma(1-\alpha)} \left(\frac{f(s)}{(s-a)^\alpha} + \alpha \int_a^s \frac{f(s) - f(u)}{(s-u)^{1+\alpha}} du \right)$$

$$D_{t-}^{1-\alpha} \omega(s) := \frac{(-1)^{1-\alpha}}{\Gamma(\alpha)} \left(\frac{\omega(s)}{(t-s)^{1-\alpha}} + (1-\alpha) \int_s^t \frac{\omega(s) - \omega(u)}{(s-u)^{2-\alpha}} du \right),$$

$0 \leq a < t \leq T$. Following Nualart and Răşcanu (2002), Zähle (1998), we define

$$\int_a^b f dw := (-1)^\alpha \int_a^b D_{a+}^\alpha f(s) D_{b-}^{1-\alpha} \omega_{b-}(s) ds,$$

where $\omega_{b-}(s) = \omega(s) - \omega(b)$.

Lemma

Assume that $\omega \in C^\nu([s, t] \times \mathbb{R})$ and $f \in C^\beta([s, t], \mathbb{R}^d)$ such that $\nu > \beta > \frac{1}{2}$. Then

$$\left\| \int_s^t f(u) d\omega(u) \right\| \leq \kappa_1 \|\omega\|_{\nu, s, t} (t-s)^\nu \left[\|f\|_{\infty, s, t} + \kappa_2 \|f\|_{\beta, s, t} (t-s)^\beta \right].$$

Lemma

For a function $Q \in C^1([a, b] \times \mathbb{R})$ such that $\frac{\partial Q}{\partial v}(\cdot, \omega(\cdot)) \in C^\nu(a, b)$ and $\alpha + \nu > 1$. Then

$$Q(t, \omega(t)) - Q(a, \omega(a)) = \int_a^t \frac{\partial Q}{\partial u}(s, \omega(s)) ds + \int_a^t \frac{\partial Q}{\partial v}(s, \omega(s)) d\omega(s).$$

For any function $\eta \in C^{1-\alpha}([a, b], \mathbb{R}^d)$ and any partition $\mathcal{P}_{t,\delta} : 0 = s_0 < s_1 < \dots < s_n = t$ of the interval $[a, b]$ such that $\sup_i s_{i+1} - s_i = \delta$, consider the step function

$$\eta_{\mathcal{P}}(s) := \sum_{i=0}^{n-1} \eta(s_i) 1_{[s_i, s_{i+1})}(s).$$

According to Zaehele (1998),

$$\lim_{\delta \rightarrow 0} \sup_{\mathcal{P}_{t,\delta}} \|\eta_{\mathcal{P}} - \eta\|_{L^\infty(0,t)} = 0$$

$$\lim_{\delta \rightarrow 0} \sup_{\mathcal{P}_{t,\delta}} \|D_{0^+,t-}^\alpha(\eta_{\mathcal{P}}) - D_{0^+,t-}^\alpha(\eta)\|_{L^1(0,t)} = 0,$$

thus the fractional integral of η can be approximated by the one of step functions, i.e.

$$\int_0^t \eta(s) d\omega(s) = \lim_{\delta \rightarrow 0} \sum_{i=0}^{n-1} \eta(s_i) [\omega(s_{i+1}) - \omega(s_i)].$$

Existence and uniqueness of solutions: fSDE

Stochastic differential equations (SDE) with fractional noise:
Nualart and Rascanu (2002)

$$\begin{aligned}dX_t &= F(t, X_t)dt + G(t, X_t)dB^H(t), \quad t \geq 0, H \in (0, 1) \\ X_0 &= x \in \mathcal{H}.\end{aligned}$$

Conditions:

- ▶ F, G are Lipschitz continuous w.r.t. x
- ▶ G is continuously differentiable with a globally Hoelder derivative. Also G, DG are Hoelder-continuous w.r.t. t .
- ▶ F, G satisfies linear growth rate condition.

Generation of random dynamical systems (RDS)

X - Banach space. RDS is a measurable mapping

$\varphi : \mathbb{R}^+ \times \Omega \times X \rightarrow X$ satisfying the cocycle property

$$\begin{aligned}\varphi(t+s, \omega, x) &= \varphi(t, \theta_s \omega, \cdot) \circ \varphi(s, \omega, x) \quad \forall t, s \in \mathbb{R}_+, \omega \in \Omega, x \in X \\ \varphi(0, \omega) &= \text{id}_X.\end{aligned}$$

Results:

- ▶ Arnold (1998), Kloeden (1994), Crauel (1994), Schmalfuss (1998, 2000, 2002): SDE or SPDE with time independent coefficients F, G generate RDS
- ▶ G.Atienza, Schmalfuss (2010, 2014): fSDE, fSPDE with time independent coefficient generate RDS.
- ▶ Mohammed, Scheutzov (2003): SDDE generate RDS on the product space. Counter-example $dX(t) = X(t-r)dB(t)$ for RDS on function space.
- ▶ Duc et al. (2014), G.Atienza (2014): fSDDE generate RDS.

Conjugacy transformation: Doss-Sussmann technique

- ▶ Sussmann (1978): there exists an explicit transformation which transforms

$$dX_t = F(X_t)dt + G(X_t)dB(t) \text{ into} \quad (1)$$

$$dY_t = f(Y_t) + \sigma dB(t) \quad (2)$$

- ▶ Imkeller-Schmalalfuss (1998), Imkeller-Lederer(2001): there exists an explicit transformation which transforms (1) into a random differential equation

$$\frac{dY_t}{dt} = f(\theta_t\omega, Y_t) \quad (3)$$

- ▶ Mohammed-Scheutzov (2003): there exists an explicit transformationn which transform

$$dX_t = F(X_t)dt + G(X_t)dB(t) \text{ into (3).} \quad (4)$$

- ▶ Question: Does there exist an explicit transformation for

$$dX_t = F(X_t)dt + G(X_t)dB^H(t) \text{ into (3) or}$$

Theorem

Under the regularity conditions, there exists a transformation which transform

$$dX_t = F(X_t)dt + G(X_t)dB^H(t)$$

into a random differential equation.

Sketch of the proof: $z(t, 0, z_0)$ - the solution of the autonomous differential equation $\dot{z} = G(z)$ generates a C^1 semigroup $\varphi(\cdot, \cdot) : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ defined by $\varphi(t, z_0) = z(t, 0, z_0)$. It also follows from the autonomy that

$$\frac{d\varphi}{dz}(t, \varphi(t, z_0))G(z_0) = G(\varphi(t, z_0)). \quad (5)$$

It can be proved that

$$\varphi(\omega(t), z_0) - z_0 = \int_0^t \frac{d\varphi}{ds}(\omega(s), z_0) d\omega(s) = \int_0^t G(\varphi(\omega(s), z_0)) d\omega(s).$$

Hence, $\varphi(\omega(t), z_0)$ is the solution of the fractional SDE $dy(t) = G(y(t))d\omega(t)$. We introduce the following transformation $X(t) = \varphi(\omega(t), u(t))$. Then we can prove that

$$u(t) - u(0) = \int_0^t \frac{d\varphi}{dx}(-\omega(s), X_s) F(X_s) ds,$$

where the last equality is due to (5). This proves that $u(\cdot)$ is differentiable w.r.t. t and satisfies the following random differential delay equation

Consider for parameter $\mu > 0$ the fractional Langevin equation

$$dz(t) = -\mu z(t)dt + dB^H(t). \quad (7)$$

Lemma

Equation (7) possesses a solution which can be computed explicitly as

$$z_\mu(t, \omega) = B^H(t, \omega) - \mu e^{-\mu t} \int_{-\infty}^t e^{\mu u} B^H(u, \omega) du, \quad t \in \mathbb{R}. \quad (8)$$

The solution z_μ is stationary, i.e. $z_\mu(t, \omega) = z_\mu(0, \theta_t \omega)$. Moreover, $E z_\mu(0, \cdot)^2 < C \mu^{-2H}$ for some constant $C > 0$ independent of μ , in particular there exists a subsequence of parameters $\mu_n \rightarrow \infty$ such that $z_{\mu_n}(0, \omega) \rightarrow 0$ as $n \rightarrow \infty$ a.s.

Theorem

Under the regularity conditions of coefficient functions, there exists a transformation which transform

$$dX_t = F(X_t)dt + G(X_t)dB^H(t) \quad (9)$$

into a random differential delay equation which is conjugated to (9).

Sketch of the proof: First, denote $D(\mathbb{R}^d)$ the Polish space of diffeomorphisms of $\mathbb{R}^d$. Then $\psi(0, \omega) := \varphi(z_\mu(0, \omega), \cdot) \in D(\mathbb{R}^d)$ and $\psi(t, \omega) := \varphi(z_\mu(t, \omega), \cdot) \in D(\mathbb{R}^d)$ satisfies the $D(\mathbb{R}^d)$ -valued SDE $d\psi(t) = G(\psi(t))dz_\mu(t)$.

Next, introduce the transformation $x(t, \omega) = \psi(t, \omega)y(t, \omega)$.

By the same technique, we can prove that $y(t, \omega)$ is a differentiable function w.r.t. t which satisfies the transformed system of the form

$$\begin{aligned} dy(t) &= g(\theta_t \omega, y_t) dt, \text{ where} \\ g(\omega, \eta) &= \left(\frac{\partial \psi}{\partial x} \right)^{-1}(\omega) [F(\psi(\omega)\eta) + \mu G(\psi(\omega)\eta) z_\mu(0, \omega)]. \end{aligned} \tag{10}$$

Finally, let $\kappa_1(t, \omega), \kappa_2(t, \omega)$ denote the solution of (9) and (10).

It is easy to check from the construction that

$$\kappa_1(t, \omega) = \psi(\theta_t \omega) \circ \kappa_2(t, \omega) \circ \psi(\omega)^{-1}.$$

Example

Consider the following scalar fSDE

$$dx(t) = \left[ax(t) + f(x(t)) \right] dt + \sigma x(t) dB^H(t). \quad (11)$$

where $\alpha, \sigma \in \mathbb{R}$ are parameters, f is a smooth function satisfying

$$\begin{aligned} xf(x) &\leq c + bx^2 \text{ for all } x \in \mathbb{R}, \text{ where } a, b, c \in \mathbb{R}, a + b < 0 \leq c \\ (x - y) \left[f(x) - f(y) \right] &\leq -L(x - y)^2 \text{ for all } x \in \mathbb{R}, \text{ where } L > a. \end{aligned}$$

Notice that a similar system for standard Brownian motion, i.e. $H = \frac{1}{2}$, is presented in Arnold [?]. The transformed system from (11) has the form

$$\dot{y}(t) = \left[a + \mu \sigma z_\mu(t, \omega) \right] y(t) + e^{-\sigma z_\mu(t, \omega)} f(e^{\sigma z_\mu(t, \omega)} y(t)). \quad (12)$$

Thus there exist a global random attractor for (11).

THANK YOU!